

the industry. There are schemes for putting semiconductor fabs in place, schemes for putting compound semiconductor fabs, schemes for lower-end fabs as well as assembly and test, and, finally, schemes to work towards increasing design capacity within the country.

EDA for design activity

In the field of electronic design automation (EDA), there is an encouraging trend for India. In the last ten years worldwide, the spending on EDA has been increasing at about 9%. In India it has increased at 12% CAGR. This indicates that design activity for the global semiconductor industry, which has been done largely within India, also provides a good platform on which the future semiconductors can be built.

Semiconductors are becoming a critical part of implementing not just near-normal electronics but also the companies' infrastructure and how they present themselves to the world. The three pillars for enabling digitalisation for outsized growth are:

Technology scaling. One of the fundamental benefits of people implementing this digitalisation is the fact that we can continue with Moore's Law, scale technology, and deliver more functionality at the same cost over time, which enables them to transform themselves and have a very positive future.

Moore's Law also has a dark side. Twice as many transistors can be produced every two years which will, in turn, require twice as many transistors to be used as well. Hence, we need to put new methodologies in place that enable us to do so.

System scaling. The application needs to get validated across the semiconductors before these are produced to ensure that overall the application can fit within the parameters it has. Also, it needs to provide the user experience it is expected to deliver in an environment where

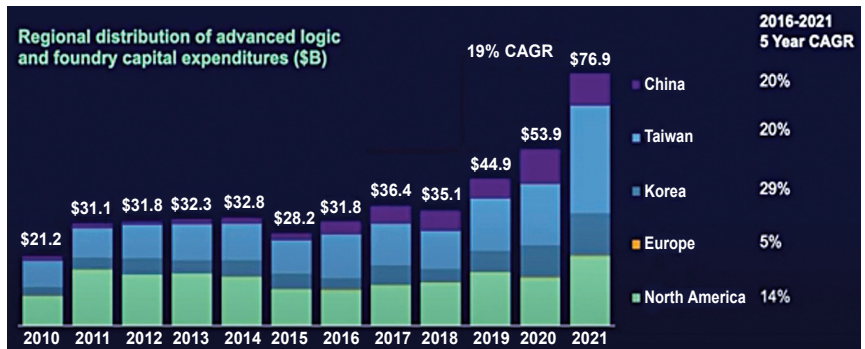


Fig. 7: Advanced logic and foundry capital investments

Semiconductor R&D spending

A rising trend can also be seen in the R&D expenditure being incurred by the established companies. Though their revenue was not increasing until 2020, that trend is now reversing. The higher revenues are translating into more R&D investment at the same time.

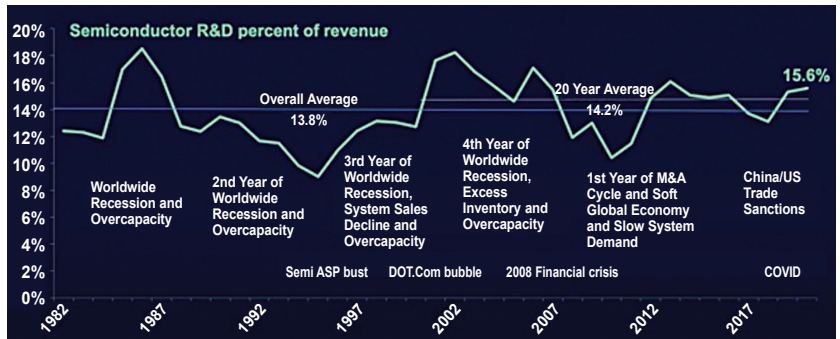


Fig. 8: Semiconductor R&D spending

	A7	A8	A9	A10	A11	A12	A13	A14	A15
Year	2013	2014	2015	2016	2017	2018	2019	2020	2021
Phone	5s	6	6s	7	8 and X	Xs and Xr	11 & SE	12 Pro Max	13 Pro Max
Technology	28nm	20nm	14/16nm	16nm	10nm	7nm	7nm N7P	5nm	5nm N5P
Die Size	102mm ²	89mm ²	94mm ²	125mm ²	88mm ²	83mm ²	98mm ²	88mm ²	107.7mm ²
# of Transistors	>1 Billion	≈2 Billion	>2 Billion	3.3 Billion	4.3 Billion	6.9 Billion	8.5 Billion	11.8 Billion	15 Billion
CPU Performance (Geekbench single core)	269	333	557	755	929	1116	1331	1589	1734
CPU Performance (Geekbench multi-core)	492 (2 cores)	524 (2 cores)	1031 (2 cores)	1420 (2 cores)	2558 (6 cores, core neural engine)	2912 (6 cores, 8 neural network)	3433 (6 cores, 8 neural cores)	4250 (6 cores, plus 16 neural cores)	4818 (6 cores, plus 16 neural cores)
GPU Performance (GFXBench 2.7 & GFX 5.0 Tess off screen)	28.7	42.8	54.6	77.3	102.5	167.2	262.4	278.3	414.1

Fig. 9: Apple's domain-specific processor evaluation

there is a sensor driving an edge computer that would work through 5G to the data centre.

Enabling technology scaling

The chart in Fig. 9 for Apple's domain-specific processor evaluation

shows an eight-year time horizon, which is equivalent to four Moore's Law cycles. It shows that in the year 2013, the A7 processor had approximately a billion transistors while in 2021 the A15 had approximately 15 billion transistors. It is